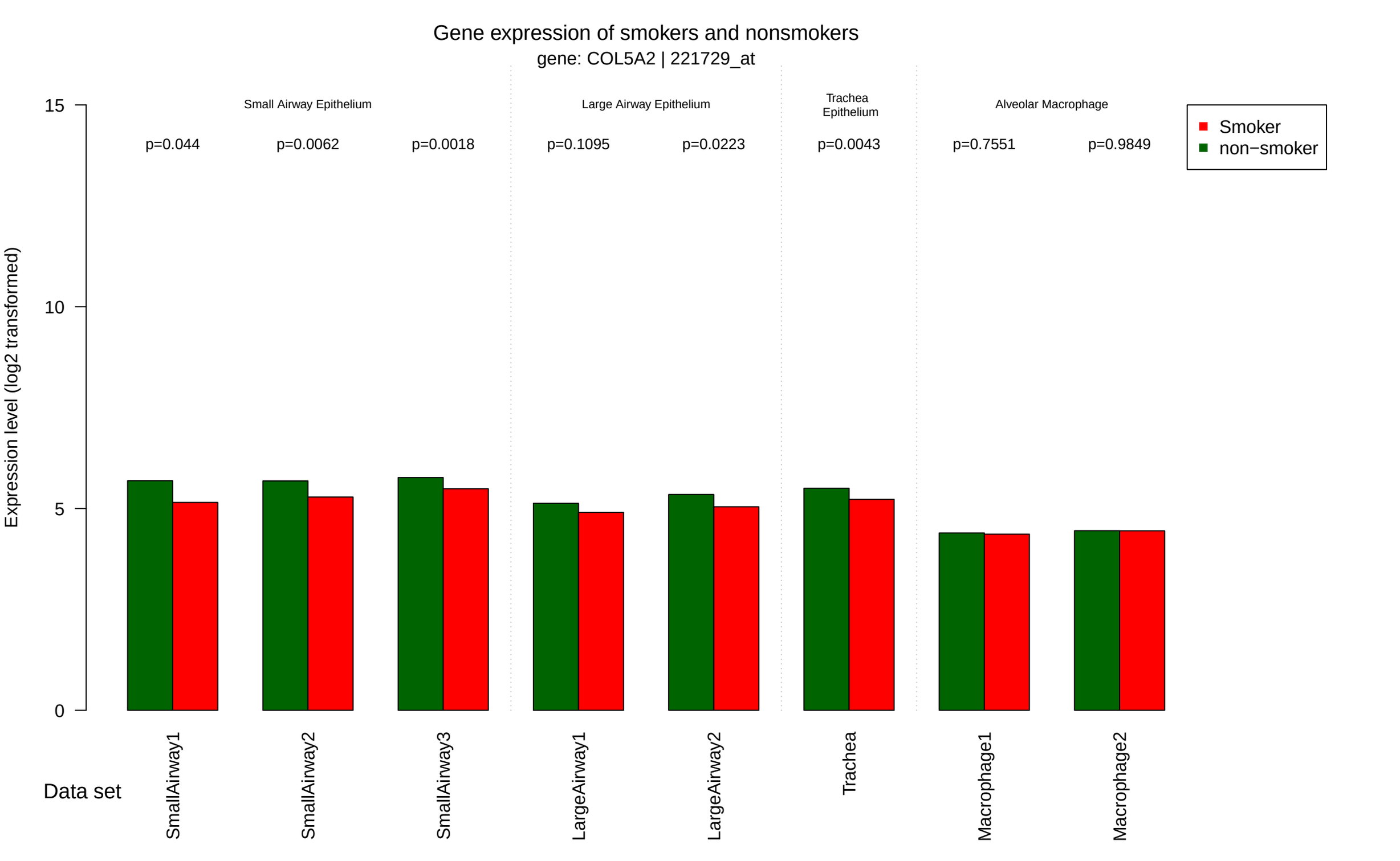
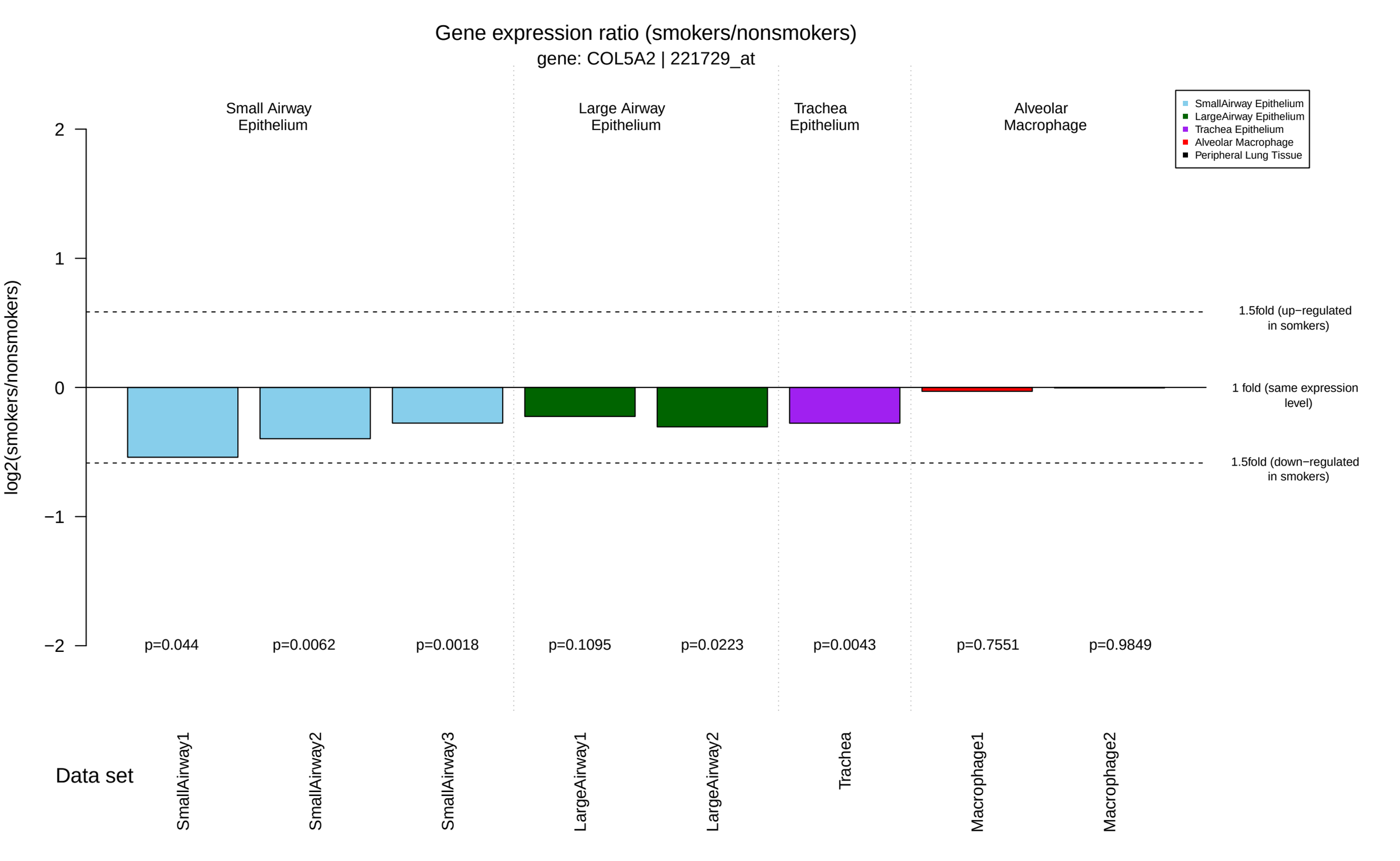
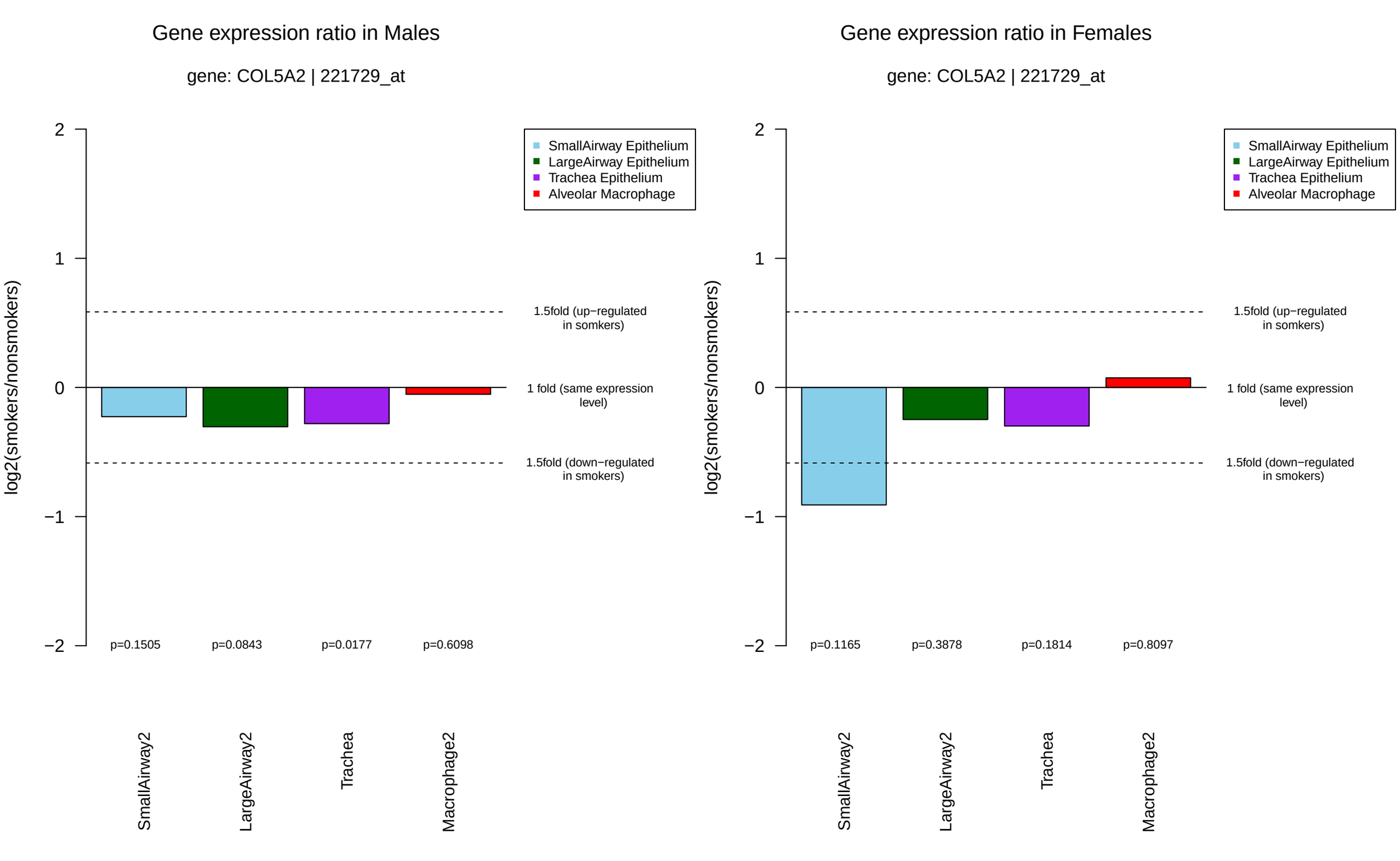
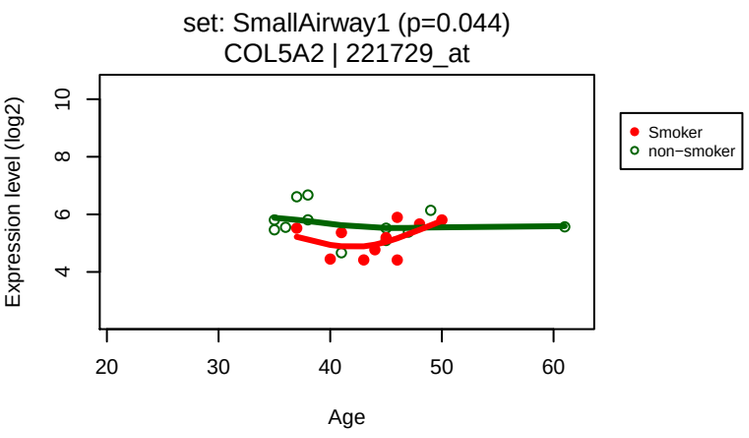
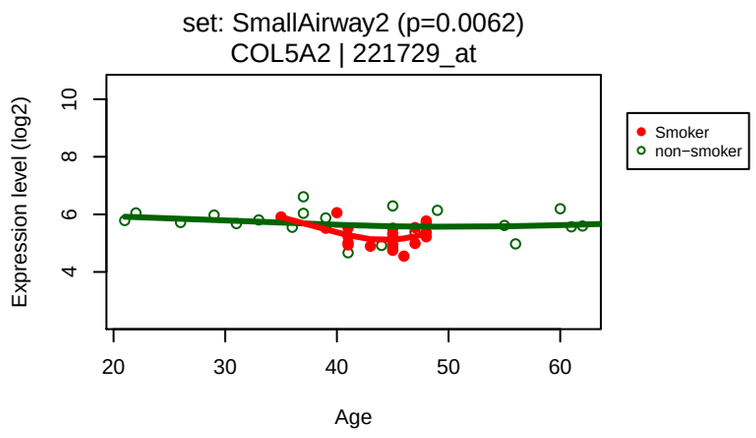
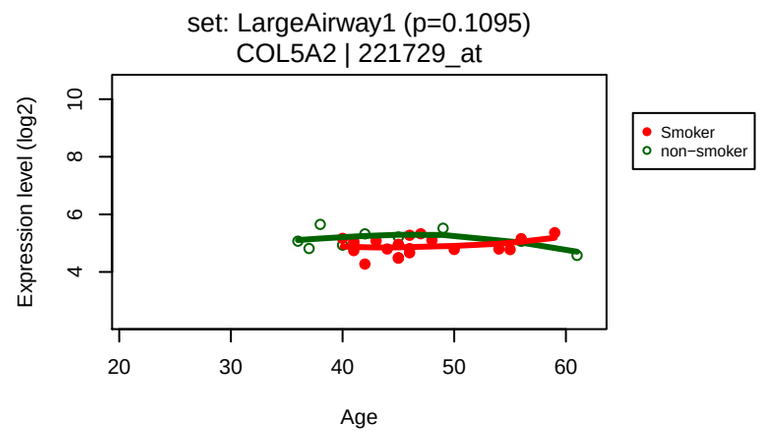

Gene expression of smokers and nonsmokers
gene: COL5A2 | 221729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.044
p=0.0062
p=0.0018
p=0.1095
p=0.0223
p=0.0043
p=0.7551
p=0.9849
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

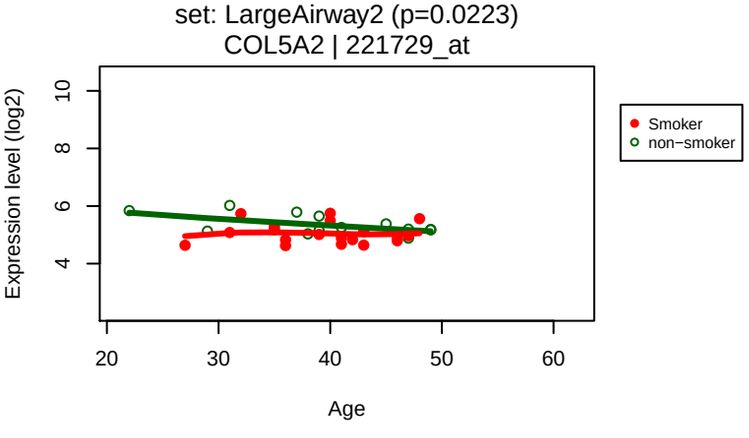

Gene expression ratio (smokers/nonsmokers)
gene: COL5A2 | 221729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.044
p=0.0062
p=0.0018
p=0.1095
p=0.0223
p=0.0043
p=0.7551
p=0.9849
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

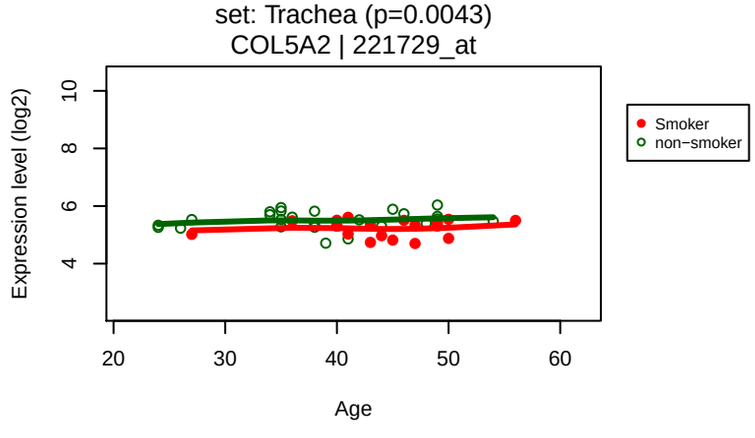

Gene expression ratio in Males
gene: COL5A2 | 221729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1505
p=0.0843
p=0.0177
p=0.6098
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL5A2 | 221729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1165
p=0.3878
p=0.1814
p=0.8097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

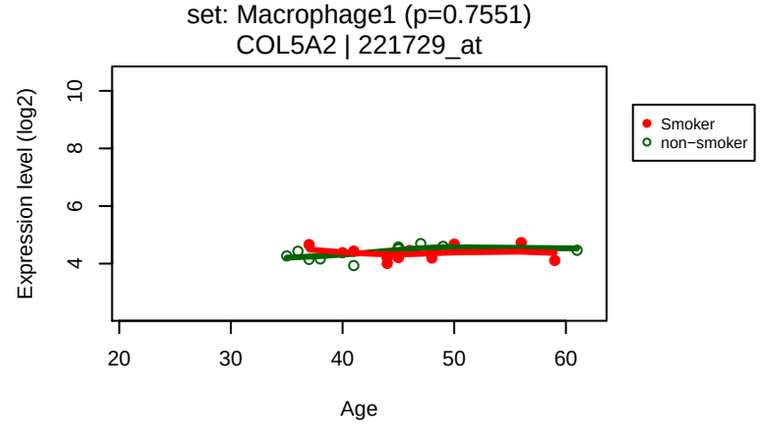

set: SmallAirway1 (p=0.044)
COL5A2 | 221729_at
Expression level (log2)
Age
Smoker
non−smoker

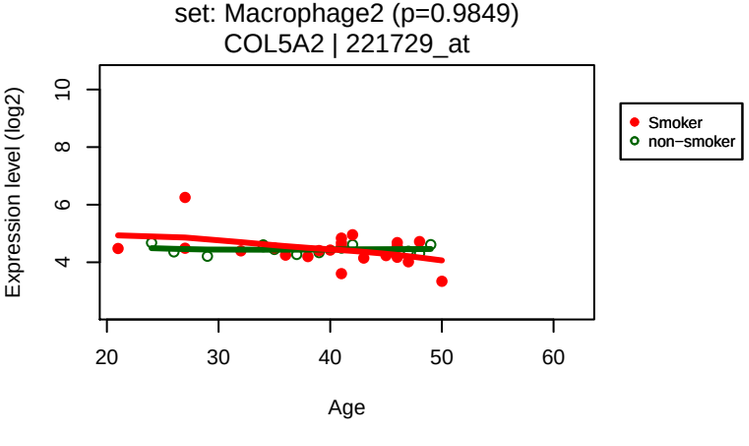

set: SmallAirway2 (p=0.0062)
COL5A2 | 221729_at
Expression level (log2)
Age
Smoker
non−smoker

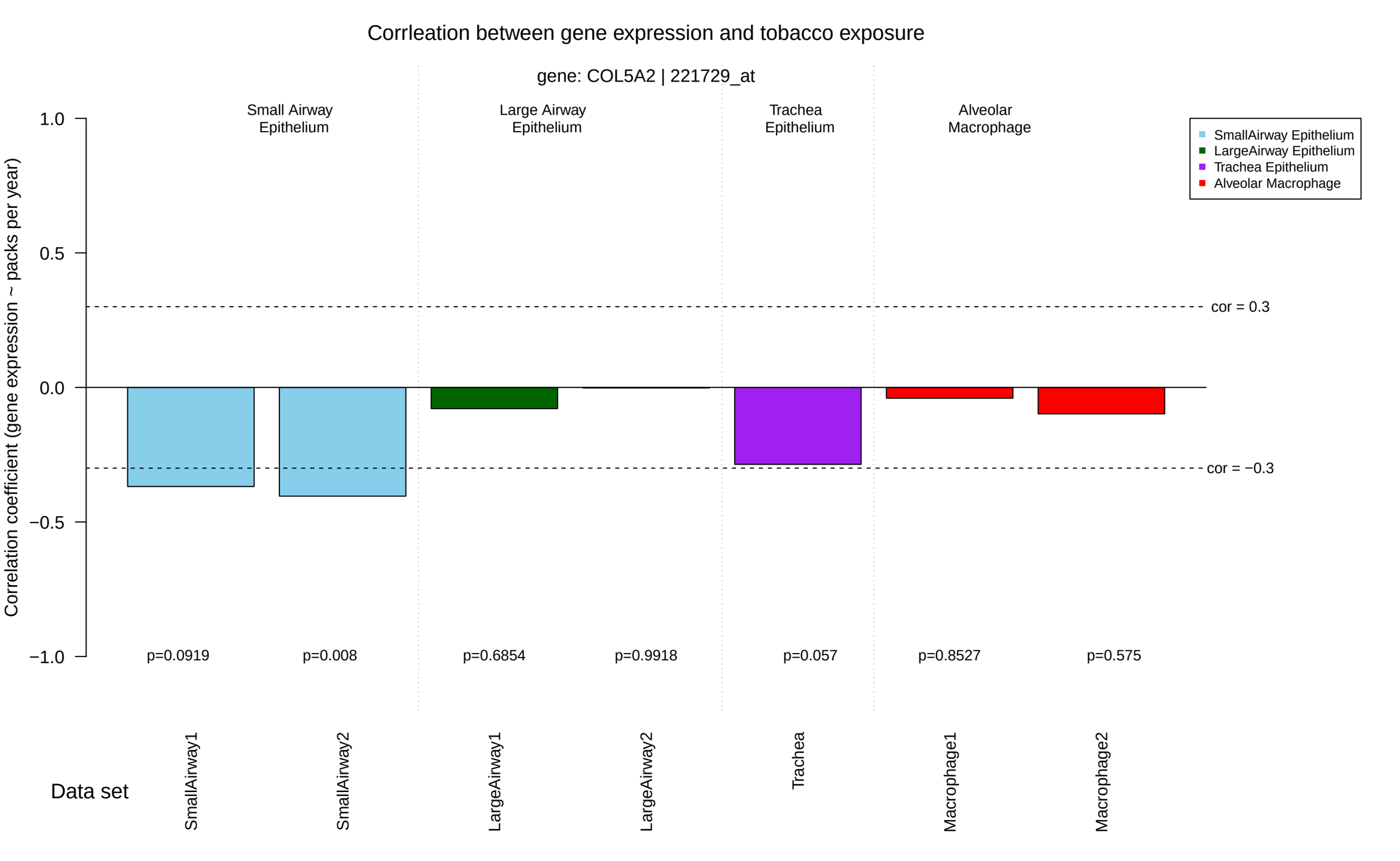

set: LargeAirway1 (p=0.1095)
COL5A2 | 221729_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0223)
COL5A2 | 221729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0043)
COL5A2 | 221729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7551)
COL5A2 | 221729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9849)
COL5A2 | 221729_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL5A2 | 221729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0919
p=0.008
p=0.6854
p=0.9918
p=0.057
p=0.8527
p=0.575
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2